Project Executable Files

Date	5 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	YES
2	README / Setup Guide (instructions to run the project)	YES
3	requirements.txt / package.json / dependency file	YES
4	Database schema / seed files (if applicable)	YES
5	Environment configuration file (.env.example or similar)	YES
6	Deployed application URL (if hosted)	YES
7	APK / Executable binary (if applicable for mobile/desktop apps)	YES
8	Dockerfile / Containerization config (if applicable)	YES
9	Test files and test results	YES
10	Demo video or walkthrough (if required)	YES

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
/OptiCrop

/templates - HTML templates for the web application
    home.html              - Home page
    about.html             - About the project
    signup.html            - User registration page
    signin.html            - User login page
    dashboard.html         - User dashboard
    findyourcrop.html      - Crop prediction input page
    result.html            - Prediction result page
    admin_login.html       - Administrator login page
    admin_dashboard.html   - Administrator dashboard

/static - Static resources
    /css                   - Stylesheets
    /js                    - JavaScript files
    /images                - Images used in the application

app.py                     - Main Flask application
model.py                   - Machine learning model training script
model.pkl                  - Trained machine learning model
Crop_recommendation.csv    - Dataset used for training
requirements.txt           - Python dependencies
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://opticrop-smart-agricultural-production-5g62.onrender.com/result?id=6
Login Credentials (Demo)	Username: admin Password: admin123

Platform / Hosting Provider	Flask Local Server (Future Deployment: Render / PythonAnywhere)
Repository Link	https://github.com/GUTTI09/AI-ML-Track-Project
Demo Video Link	https://drive.google.com/file/d/1UsIJvM8_VYTuCXaSOEBKLAI1-gSyu8BC/view?usp=drive_link

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Run Instructions

1. Install **Python 3.14 or later** on your system.
2. Download or clone the project folder.
3. Open the project folder in **Visual Studio Code**.
4. Open the terminal and install the required Python packages using:

```
pip install -r requirements.txt
```

5. Ensure that the trained model file (`model.pkl`) and dataset (`Crop_recommendation.csv`) are present in the project directory.
6. Run the Flask application using the following command:

```
python app.py
```

7. After the server starts successfully, open your web browser and navigate to:

```
http://127.0.0.1:5000/
```

8. The application home page will open. Users can register, log in, access the dashboard, enter crop parameters, and obtain crop recommendations.
9. Administrators can log in using the **Admin Login** page to access the admin dashboard and monitor the application.
10. To stop the application, press **Ctrl + C** in the terminal.

After running this code the following steps are involved to run render platforms

1. Open any modern web browser (Google Chrome, Microsoft Edge, or Firefox).
2. Open the deployed application using the Render URL:

```
https://your-render-app.onrender.com
```

Note: `your-render-app` place lo mee actual Render application URL pettandi.

3. The Home page will be displayed.
4. New users can create an account using the **User Registration** page.
5. Registered users can log in through the **User Login** page.
6. After successful login, navigate to the **Find Your Crop** page and enter the required soil and environmental parameters (Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall).
7. Click the **Predict** button to generate the recommended crop using the trained Machine Learning model.
8. The predicted crop will be displayed on the **Result** page.
9. Administrators can log in through the **Admin Login** page to access the **Admin Dashboard** and manage the application.
10. Log out after completing the required operations.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Predictions depend on the quality and size of the training dataset.	Can be improved by training with a larger and more diverse dataset.
2	The application currently uses an SQLite database, which is suitable for small-scale deployments.	Can be migrated to MySQL or PostgreSQL for large-scale production environments.
3	The application does not currently integrate real-time weather or IoT sensor data.	Future enhancement includes API-based weather integration and IoT support.